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Images may be on the next page

UML

Colour Key
Void
String
int
Double
Constructor

StockItem
- stockCode: String
- stockName: String
- stockDescription: String
- stockLevel: int
- vatRate: double
- stockPrice: double
+ StockItem(in: stockCode: String, stockLevel: int, stockPrice: double): constructor
+ getStockCode(): String
+ setStockCode(in: stockCode: String): void
+ getStockName(): String
+ setStockName(in: stockName: String): void
+ getStockDescription(): String
+ setStockDescription(in: stockDescription: String): void
+ getStockLevel(): int
+ setStockLevel(in: stockLevel: int): void
+ getStockPrice(): double
+ getPriceWithVAT(): double
+ setStockPrice(in: stockPrice: double): void
+ getVatRate(): double
+ setVatRate(in: vatRate: double): void
+ addStock(in: additionalStock: int): void
+ sellStock(in: unitsSold: int): void
+ toString(): String
+ consoleOutput(): String

stockList
- stockPannel: JPanel
- listOfStock: JList
- addNewStockButton: JButton
- openButton: JButton
- listOfStocklist: DefaultListModel
+ loadWindow(in: listener: ActionListener): void
+ addItemToList(in unit: StockItem): void
+ getSelectedValue(): StockItem
+ getSelectedIndex(): int
+ replaceItemAtIndex(in: index: int, unit: StockItem): void

productOverview
- productImage: JLabel
- unitsInStock: JLabel
- costWithVAT: JLabel
- productNameText: JTextField
- unitCostText: JTextField
- itemDescriptionText: JTextField
- unitsInStockText: JTextField
- vatRateText: JTextField
- updateButton: JButton
- sellItemButton: JButton
- addStockButton: JButton
- overviewPannel: JPanel
+ loadWindow(in: listener: ActionListener, unit: StockItem): void
+ getProductName(): String
+ getUnitCost(): double
+ getItemDescriptionText(): String
+ getUnitsInStock(): int
+ getVatRate(): double
+ refresh(in unit: StockItem): void

NewProductForm
- ProductCode: JTextField
- Price: JTextField
- ProductName: JTextField
- VatRate: JTextField
- UnitsOfStock: JTextField
- saveButton: JButton
- abortButton: JButton
- NewProductWindow: JPanel
+ loadWindow(in: listener: ActionListener): void
+ getVatRate(): double
+ getProductCode(): String
+ getProductPrice(): double
+ getProductStockLevel(): int
+ getProductName(): String

Controller
- stockWindow: stockList
- newProductWindow: newProductForm
- productWindow: productOverview
+ main(): void
+ actionPerformed(in: e: ActionEvent): void

ActionListener

Realization

Inheritance

Step 1
UML

Step 2
UML

NavSys
+ NavSys(in: stockCode: String, stockLevel: int, stockPrice: double): constructor
+ getStockName(): String
+ getStockDescription(): String

ActionListener

Realization

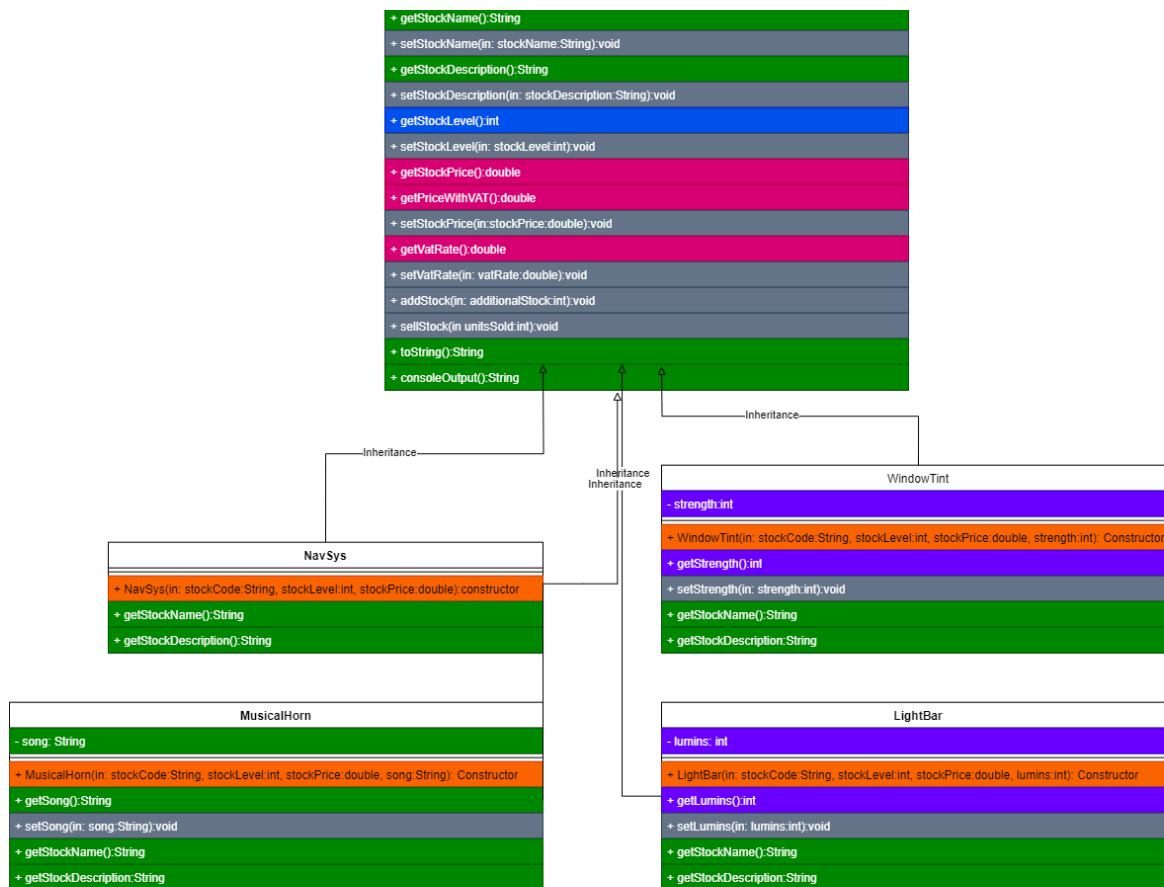
TestNavSys
- newProductWindow: newProductForm
- productWindow: productOverview
- stockWindow: stockList
+ main(): void
+ actionPerformed(in: e: ActionEvent): void

Step 2
UML

Step 3
UML

StockItem
- stockCode: String
- stockName: String
- stockDescription: String
- stockLevel: int
- vatRate: double
- stockPrice: double
+ StockItem(in: stockCode: String, stockLevel: int, stockPrice: double): constructor
+ StockItem(in: stockCode: String, stockLevel: int, stockPrice: double, vatRate: double): constructor
+ getStockCode(): String
+ setStockCode(in: stockCode: String): void

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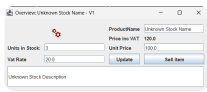
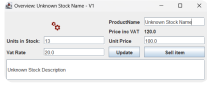


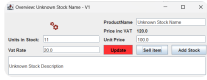
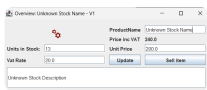
TestPolymorphism
- stockWindow: stockList
- newProductWindow: newProductForm
- productWindow: productOverview
+ main():void
+ itemInstance(in: StockItem): void

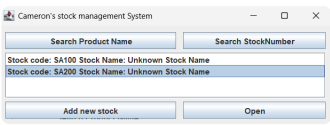
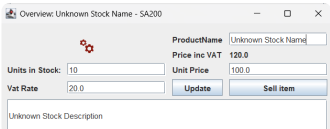
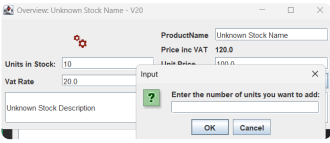
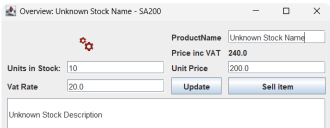
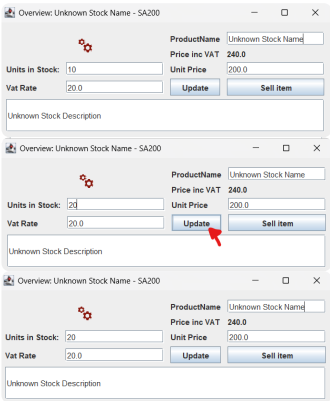
Step 3
UML

Testing

Testing from 1.2

Test Case	What will be done	Purpose	Expected Result	Actual Result	PASS/FAIL
CLASS1	Create a stock with 3 units, with a price of 100 each and a VAT of 20%	To test that the stockItem class works as expected. And the GUI class can display the data correctly	GUI will show correct stock and quantity and a price of 120 inc VAT		PASS
CLASS2	Increase the units in stock by 10	To test that the class is able to be updated.	GUI will show a value of 13 after update		PASS

Test Case	What will be done	Purpose	Expected Result	Actual Result	PASS/FAIL
CLASS3	Sell 2 units using the sell item button	To test that the units in stock are correctly reduced	GUI will show 11 units after this		FAIL PASS
CLASS4	Update unit price to be 200	To test that the GUI and stockItem can correctly handle the change.	GUI will show a unit price of 200 and a price inc VAT of 240.		PASS

Test Case	What will be done	Expected Result	Actual Result	PASS/FAIL
GUI1	Create multiple StockItem objects	List will have more than one row		PASS
GUI2	Add stock From 6 to 10	Overview will show the updated stock when updated and reopened		PASS
GUI3	Sell Stock	GUI will show popup to ask user how many units to sell.		PASS
GUI4	Change item prices From 100 to 200	Price will show 200 and price inc VAT will show 240 after updating and reopening the overview		PASS
GUI5	Display updated item details	Make any change and show it as working in the GUI. In this test I changed the stock from 10 to 20		

Testing from 3.2

Test Case	What will be done	Expected Result	Actual Result	PASS/FAIL
PlyMorph1	Decrease the stock level of the lightBar Object by 10	Stock level will be set to 0	<pre> outputting stock information: Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:10 Vat:0.0% Price: 100.0 VAT Price: 100.0 -----Units updated Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:0 Vat:0.0% Price: 100.0 VAT Price: 100.0 </pre>	PASS
PlyMorph2	Increase the stock level by 30	Stock level will be set to 30	<pre> -----Units updated Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:0 Vat:0.0% Price: 100.0 VAT Price: 100.0 -----Units Updated Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:30 Vat:0.0% Price: 100.0 VAT Price: 100.0 </pre>	PASS
PlyMorph3	Set price to 300	Price and VAT price will be set to 300	<pre> -----Price Updated Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:30 Vat:0.0% Price: 300.0 VAT Price: 300.0 </pre>	PASS
PlyMorph4	Check that the stock level change works on the next class in the array	Stock level will be set to 5	<pre> outputting stock information: Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:10 Vat:0.0% Price: 100.0 VAT Price: 100.0 -----Units updated Stock code:LB1 Stock name:Light Bar Description: Roof light bar that provides 360 Lighting, Lumins:3200 Stock level:5 Vat:0.0% Price: 100.0 VAT Price: 100.0 </pre>	PASS